Binary Tree Vertical Order Traversal (medium)

Problem Statement

Given the `root` of the binary tree, return the 2D list containing the `vertical order traversal` of the binary tree.

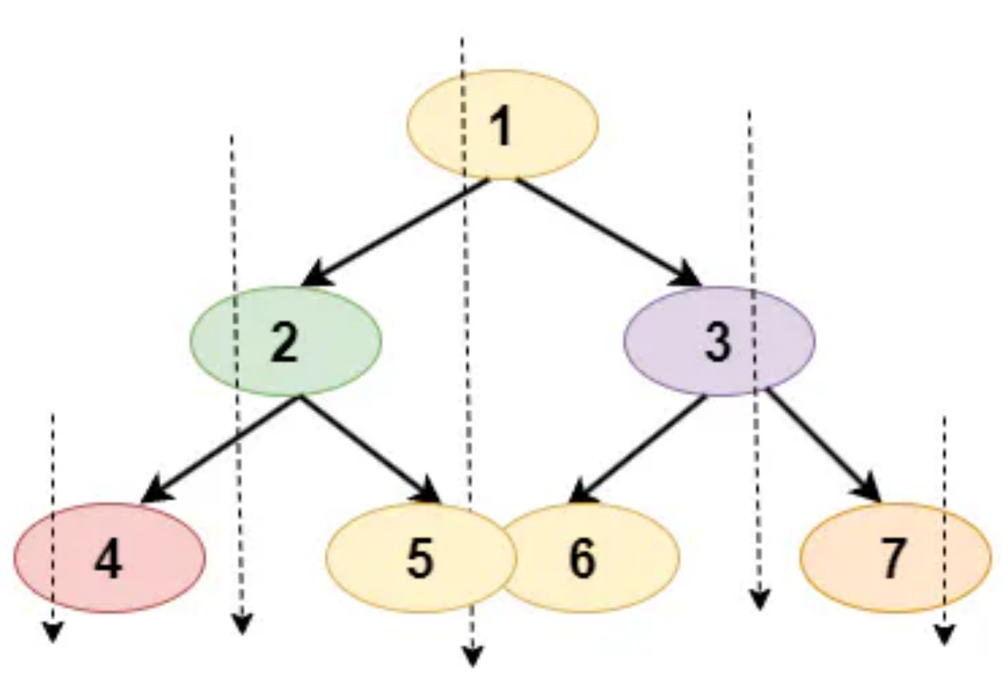
A vertical order traversal of the tree is defined as a `top to bottom, column by column` traversal.

Note: If two nodes are in the same row and column, keep its order from left to right.

Examples

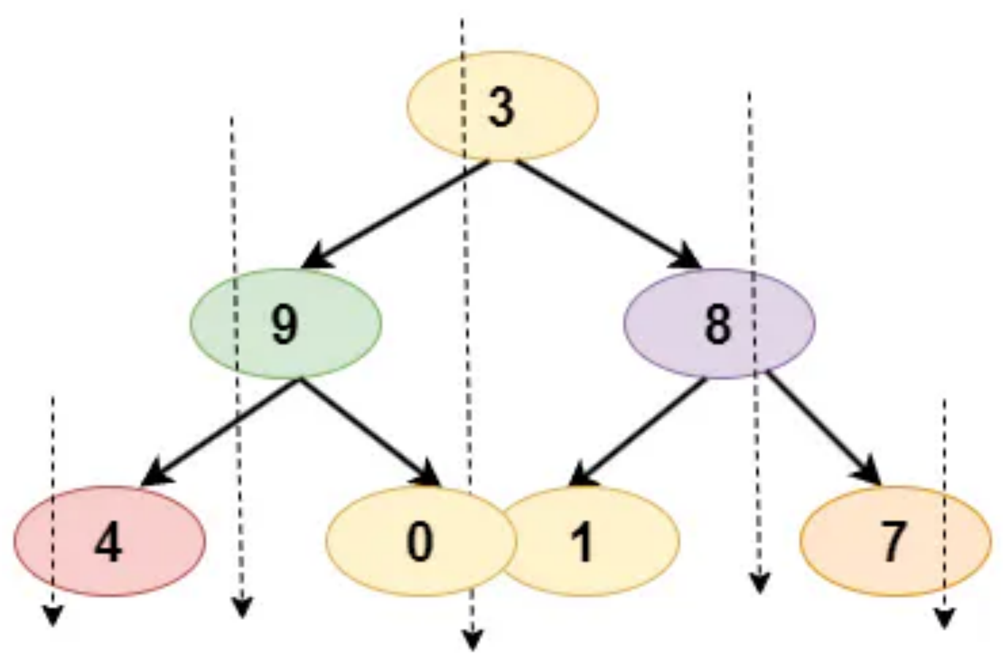
Example 1:

- **Input:** A binary tree: `[1,2,3,4,5,6,7]`
- **Expected Output:** `[[4], [2], [1,5,6], [3], [7]]`
- **Justification:** Nodes 4, 2, 1 with 5 and 6, 3, and 7 are in separate vertical lines. The nodes in each vertical line are listed in the order they appear from top to bottom.



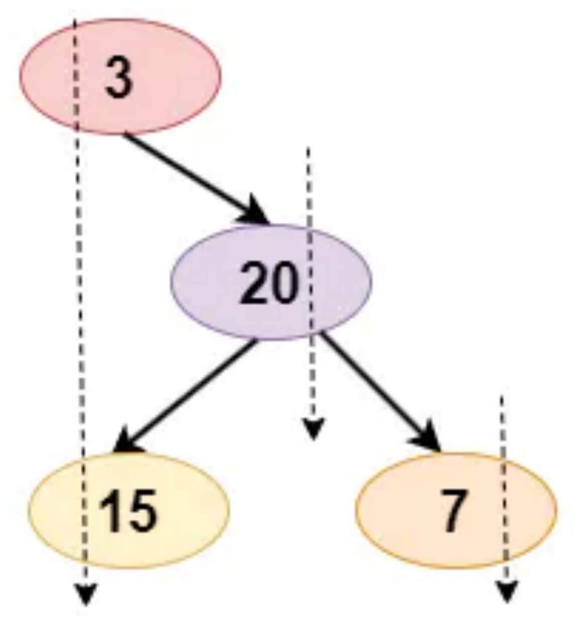
Example 2:

- **Input:** A binary tree: `[3,9,8,4,0,1,7]`
- **Expected Output:** `[[4], [9], [3,0,1], [8], [7]]`
- **Justification:** Nodes are grouped based on their vertical positions. Lower nodes in the same vertical line follow the higher ones.



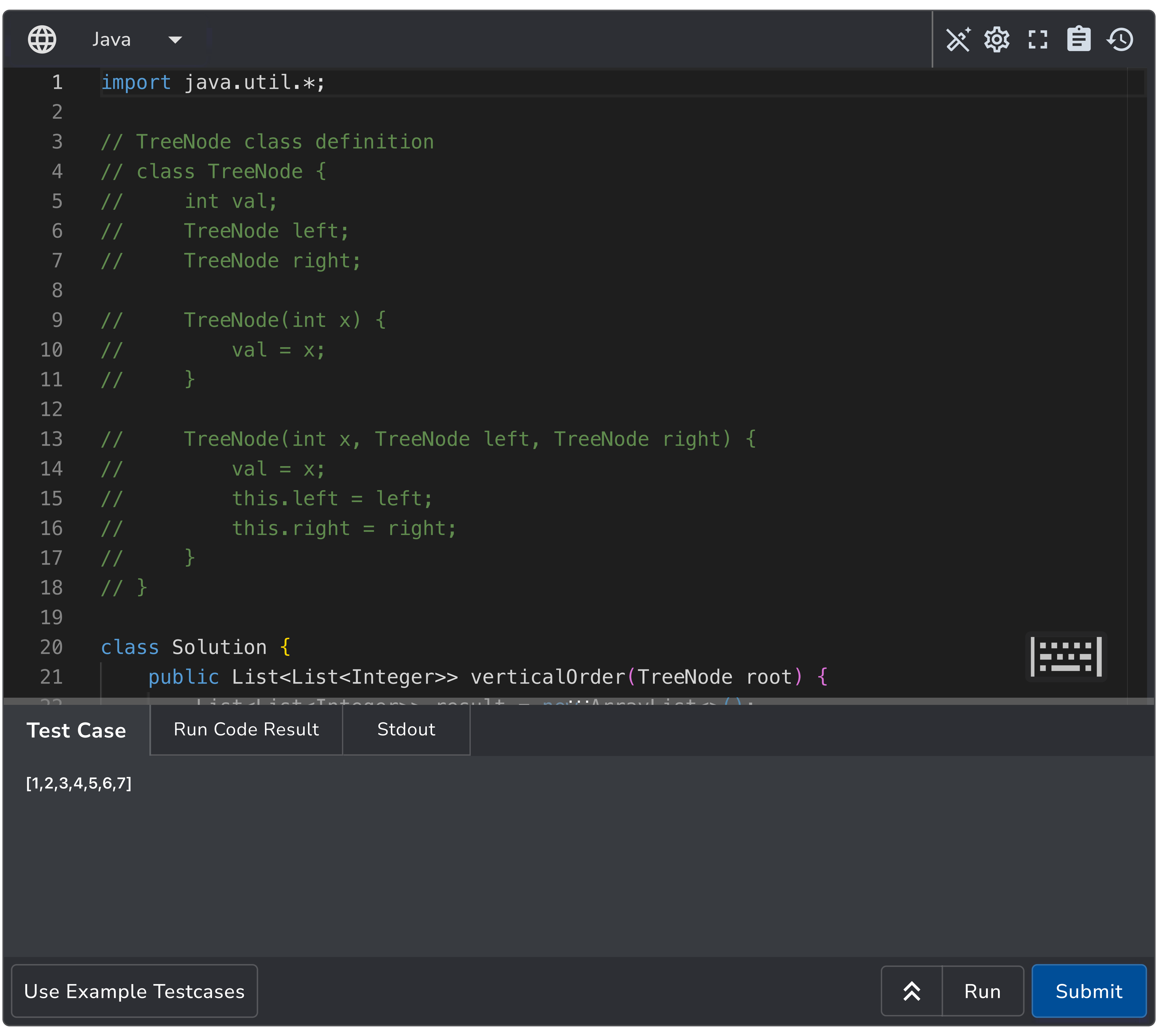
Example 3:

- **Input:** A binary tree: `[3,null,20,15,7]`
- **Expected Output:** `[[3, 15], [20], [7]]`
- **Justification:** The tree is skewed to the right. The output reflects the vertical traversal from left to right.



Try it yourself

Try solving this question here:



```java
import java.util.*;

// TreeNode class definition
// class TreeNode {
//     int val;
//     TreeNode left;
//     TreeNode right;

//     TreeNode(int x) {
//         val = x;
//     }

//     TreeNode(int x, TreeNode left, TreeNode right) {
//         val = x;
//         this.left = left;
//         this.right = right;
//     }
// }

class Solution {
    public List<List<Integer>> verticalOrder(TreeNode root) {
```

[1,2,3,4,5,6,7]

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Solution: Binary Tree Vertical Order Traversal

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